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How to build a Python DSL for MLIR

`mlir dsl` · Python generates MLIR

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How do we produce MLIR?

- Write MLIR by hand

Static text. Every variant is manual.

- Use the Python bindings

Generate IR. Wire up compilation and execution.

Just a + b, with Python bindings

1 · Build the IR

```
import ctypes as C
from mlir import ir, execution_engine as ee
from mlir import passmanager as pm
from mlir.dialects import arith, func

with ir.Context(), ir.Location.unknown():
    i32 = ir.IntegerType.get_signless(32)
    module = ir.Module.create()
    with ir.InsertionPoint(module.body):
        fn = func.FuncOp(
            "add", ([i32, i32], [i32]))
        fn.attributes["llvm.emit_c_interface"] = (
            ir.UnitAttr.get())
        entry = fn.add_entry_block()
        with ir.InsertionPoint(entry):
            a, b = fn.arguments
            result = arith.AddIOp(a, b).result
            func.ReturnOp([result])
```

2 · Compile + execute

```
pipeline = (
    "builtin.module("
    "convert-arith-to-llvm,"
    "convert-func-to-llvm,"
    "reconcile-unrealized-casts)"
)
pm.PassManager.parse(pipeline).run(
    module.operation)
engine = ee.ExecutionEngine(module)

a, b = C.c_int32(2), C.c_int32(3)
out = C.c_int32()
engine.invoke(
    "add", C.byref(a), C.byref(b),
    C.byref(out))
print(out.value) # 5
```

Complete program: build IR → lower → JIT → call.

Now add PyTorch tensors

Same CPU tensors. Compiled $a + b$.

PyTorch CPU tensors

```
import torch

a = torch.arange(4, dtype=torch.float32)
b = torch.full_like(a, 10.0)
out = torch.empty_like(a)
# Expected: tensor([10., 11., 12., 13.])
```

mlir dsl · this is what we want

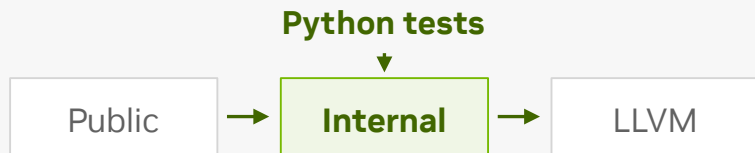
```
import mlir.mlir_dsl as m
P = m.Pointer[m.Float32]

@m.jit
def add(n: m.Int32, a: P, b: P,
        out: P):
    for i in range(n):
        out[i] = a[i] + b[i]

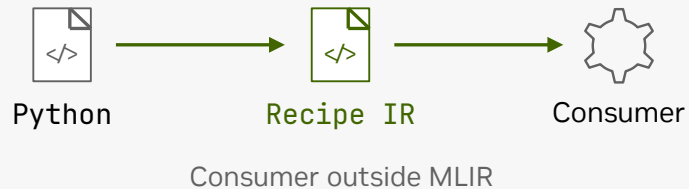
add(a.numel(), a, b, out)
print(out)
```

Four use cases inside NVIDIA

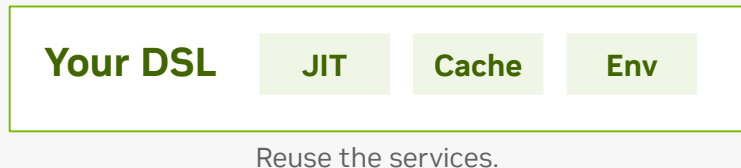
01 Test internal dialects



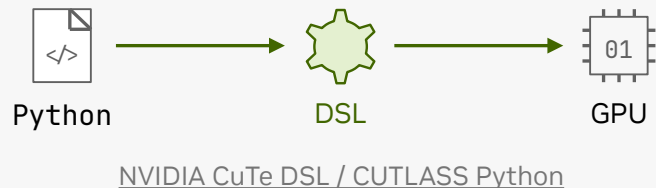
02 Generate recipe IR



03 Reuse DSL building blocks



04 Build a full language



How do we build mlir dsl?

01 Which dialects?

Every compiler has its own IR.

02 Which control flow?

Different targets need different rules.

03 Generate IR—or execute?

There is no single end-to-end pipeline.

Share the core services

Tracing

Run Python. Capture operations.

Type inference

Promotion + casts

Diagnostics

Source locations + clear errors

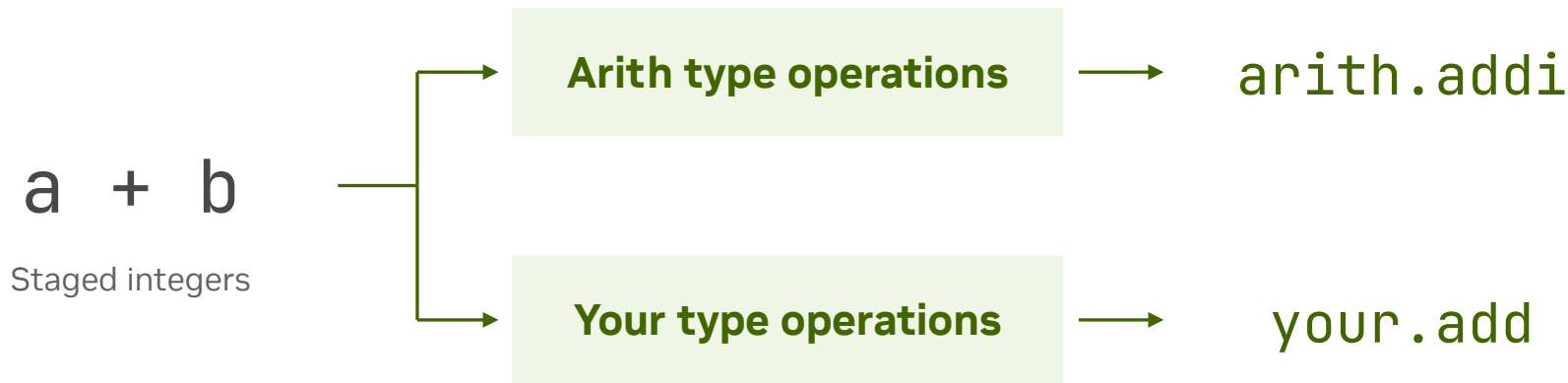
Cache

Memory + disk

Core services only. No default dialect or compiler.

What is a plugin?

A replaceable implementation of part of your DSL.



Same Python expression. Your DSL chooses the emitted IR.

Compose the DSL you need

Shared core + Selected plugins = **Your DSL**

Generate MLIR

Add compiler + your pipeline →
execute

mlir dsl is a working composition. Reuse its plugins.

How much Python should we implement?

AST frontend

Parse Python
Implement its semantics

Tracing

Run Python
Record DSL operations

Our choice: tracing. Optional AST capture handles staged control flow.

Design principles

01 No new language.

Objects · imports · libraries

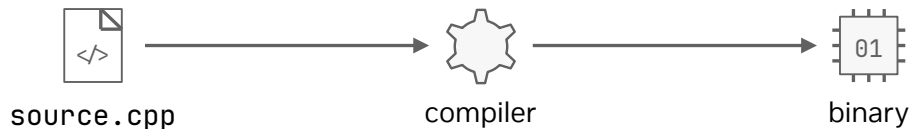
02 Python stays Python.

MLIR through a DSL namespace

This leads to **multi-stage programming**.

What is multi-stage programming?

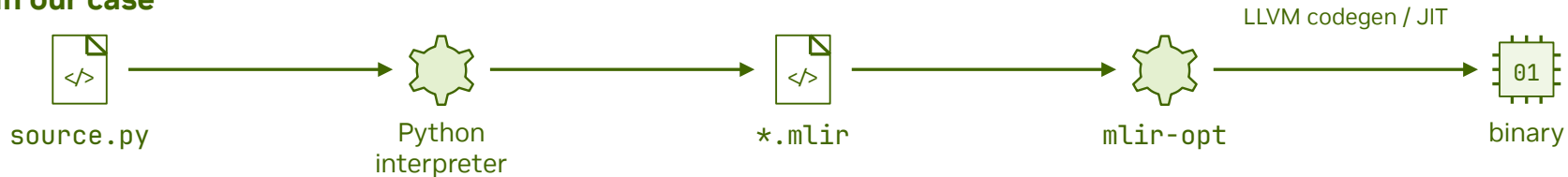
Traditional compiler



Multi-stage compilers



In our case



Why multi-stage programming?

01 METAPROGRAM

All of Python.

Objects · imports · libraries

02 DSL NAMESPACE

Explicit capture.

Types · operators · builders

```
import mlir.mlir_dsl as m
```

```
range(4)
```



Python loop

Runs now

```
m.range(4)
```



scf.for

Runs later

Agenda

- 01 Trace Python → types → control flow
- 02 Host boundary → kernels → compilation
- 03 Tools → build your own DSL
- 04 Amir: advanced mutation
- 05 Tile IR: a target composition

1 · Tracing JIT

User program

```
import mlir.mlir_dsl as m

@m.jit
def scale(x: m.Int32, factor):
    return x * factor

int(scale(4, 3))  # 12
```

Reference execution path

```
Python call
  ↓
Trace func.func
  ↓
Compile + cache
  ↓
Invoke
```

2 · Python value or MLIR value?

Python

```
@m.jit
def scale(x: m.Int32, factor):
    assert m.is_mlir_op(x)
    assert not m.is_mlir_op(factor)
    return x * factor

scale(4, 3)
```

Generated MLIR · excerpt

```
# At trace time
factor = 3

# In the generated program
%x : i32
%c3 = arith.constant 3 : i32
%r = arith.muli %x, %c3 : i32
```

Python is the metaprogram

Python · implementation excerpt

```
@m.jit
def affine(x: m.Int32, options):
    scale = options["scale"]
    bias = options["bias"]
    return x * scale + bias

options = {"scale": 3, "bias": 4}
affine(7, options) # Int32(25)
```

Runs in Python

Dictionary lookups

Captured as MLIR

Multiply + add

Objects + dictionaries → arithmetic

Python polymorphism

```
class Op:
    def apply(self, x):
        raise NotImplementedError

class AddN(Op):
    def __init__(self, n): self.n = n
    def apply(self, x): return x + self.n

class MulN(Op):
    def __init__(self, k): self.k = k
    def apply(self, x): return x * self.k
```

Metaprogram

```
@m.jit
def pipeline(x: m.Int32, ops):
    for op in ops.values():
        x = op.apply(x)
    return x

ops = {"shift": AddN(2), "scale": MulN(3)}
pipeline(4, ops) # 18

# Captured: (x + 2) * 3
```

No runtime dictionary or virtual dispatch.

3 · Numeric types are core policy

Family	Types
Integers	Int2–Int128 · Uint8–Uint128
Floating point	Float16/32/64 · BFloat16 · TFloat32
Narrow floats	FP8 · FP6 · FP4 formats
Boolean	Boolean

Literal → promotion → cast

Boolean · i1

Int2-128 / UInt8-128

FP64/32/16 · BF16

FP8/6/4 · TF32

Python or SSA value → dtype + promotion → type_ops plugin

Conversion

Python

MLIR

Capture

`m.Int32(7) → staged use`

`arith.constant : i32`

Promote / upcast

`x32 + y64`

`extsi → addi : i64`

Downcast

`y64.to(m.Int32)`

`arith.trunci`

Int → float

`x32.to(m.Float32)`

`arith.sitofp`

Float → int

`f32.to(m.Int32)`

`arith.fptosi`

Error

`x32.to(int)`

Runtime ≠ Python value

Shown: TypeOps with arith

The same dtype can target another dialect

Type-ops plugin · implementation excerpt

```
class TileTypes(TypeOps):
    def mlir_type(self, dtype):
        return ir.RankedTensorType.get(
            [], dtype.scalar_mlir_type)

class TileLike(m.MlirTestDSL):
    plugins = replace(
        m.MlirTestDSL.plugins,
        type_ops=TileTypes(scalars=arith))
```

Same source

m.Int32(6) + x

Different IR type

i32 → tensor<i32>

4 · Vectors, pointers, structs

Value	How it maps
Vector	vector hooks: create · extract · reduce
Pointer[T]	memory hooks: load · store · offset
@struct	Frozen record → flattened leaves
Custom type	register_leaf → your SSA value

Records flatten into their fields

Dialect-neutral record

```
@m.struct
class Vec2:
    x: m.Float32
    y: m.Float32

@m.jit
def norm2(v: Vec2):
    return v.x*v.x + v.y*v.y
```

Two fields → two SSA arguments

```
func.func @norm2(
    %x: f32, %y: f32) → f32 {
    %a = arith.mulf %x, %x : f32
    %b = arith.mulf %y, %y : f32
    %r = arith.addf %a, %b : f32
    func.return %r : f32
}

// norm2(Vec2(x=3, y=4)) → 25
```

Frozen record · no aggregate SSA type

Extend values with mydialect.Tensor

Custom leaf · implementation excerpt

```
@dataclass(frozen=True)
class Tensor:
    value: ir.Value

register_leaf(
    Tensor,
    ir_types=lambda t: [t.value.type],
    ir_values=lambda t: [t.value],
    from_ir_values=lambda t, vs: Tensor(vs[0]),
)
```

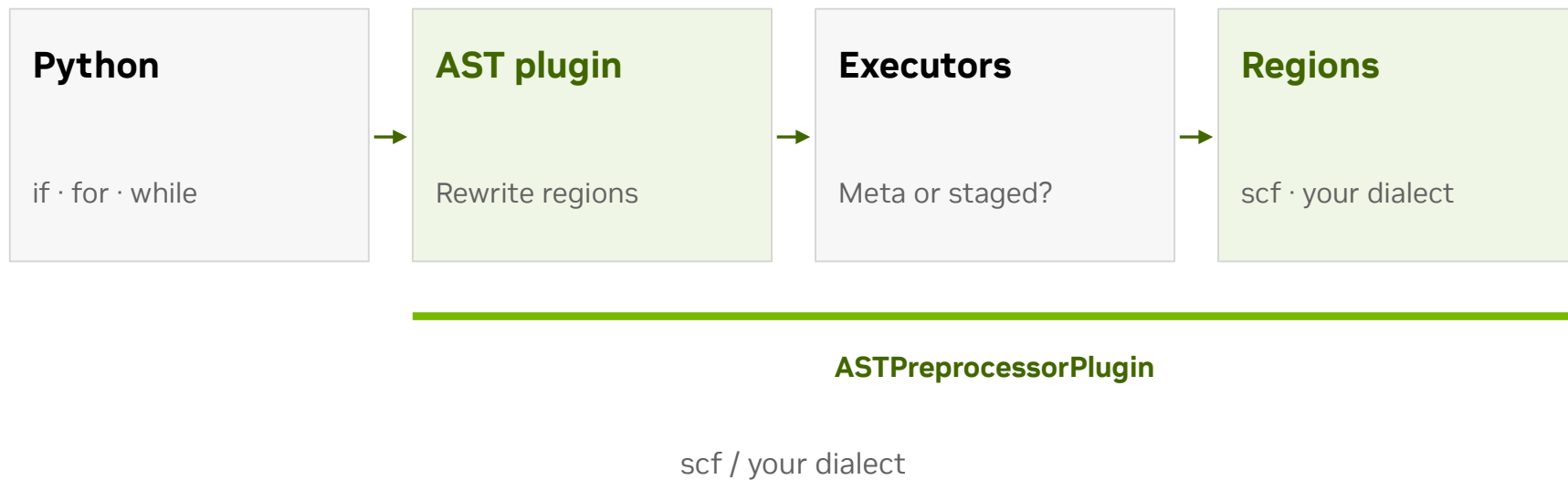
Keep a custom value

Own the Python wrapper

Cross DSL boundaries

Extract + rebuild SSA leaves

5 · Native control flow



Syntax capture and IR emission are separate choices.

Assigned name → iter_arg

Python · find the carry

```
acc = m.Int32(0)
for i in range(n):
    acc = acc + i

# Defined outside: acc
# Written inside:  acc
# write_args_names = ["acc"]
```

MLIR · thread the SSA value

```
%sum = scf.for %i = %c0 to %n
      step %c1 iter_args(%acc = %zero)
      → (i32) : i32 {
    %next = arith.addi %acc, %i : i32
    scf.yield %next : i32
  }
```

Flatten → build region → yield → rebuild

Capture has an explicit boundary

Python / Meta	Staged
if / for / while	scf.if / scf.for / scf.while
Known early exit	Python executes it
Data-dependent early exit	UNSUP_EARLY_EXIT
Frozen dataclass	Flatten + rebuild leaves

Replace capture through one role

AST plugin · implementation excerpt

```
class MyAST(scf.ASTPreprocessor):
    def executors(self, dsl):
        hooks = super().executors(dsl)
        hooks["loop_execute_range_dynamic"] = (
            my_loop_executor)
        return hooks

# In the Plugins record:
ast_preprocessor=MyAST()
```

Reuse the rewrite

Replace a region builder

Or replace both

preprocessor_class + executors

6 · Explicit builders

Python · implementation excerpt

```
@m.jit(preprocess=False)
def sum_squares(n: m.Int32):
    for i, acc, result in m.for_(
        0, n, 1, [m.Int32(0)]):
        m.yield_(acc + i * i)
    return result
```

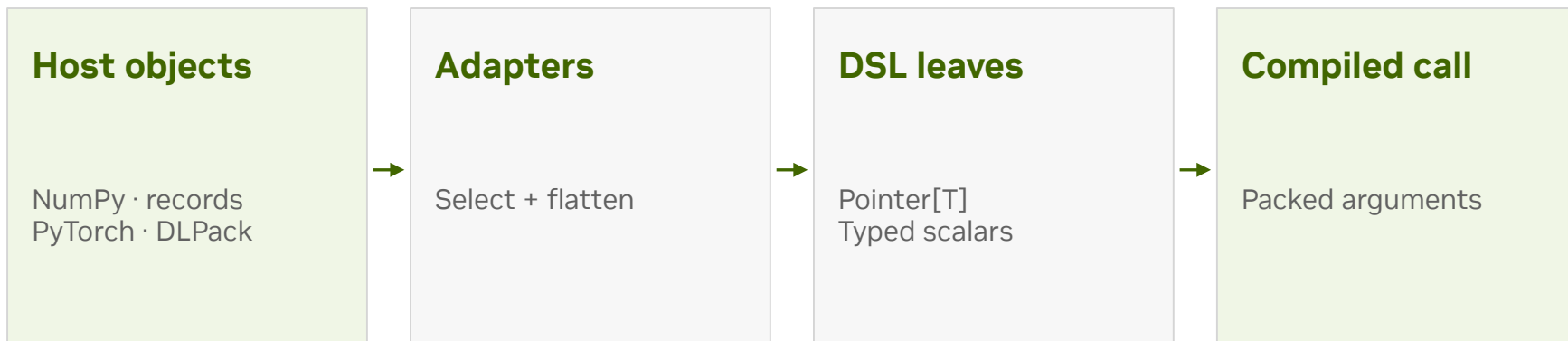
No syntax rewrite

Build regions directly

Same building blocks

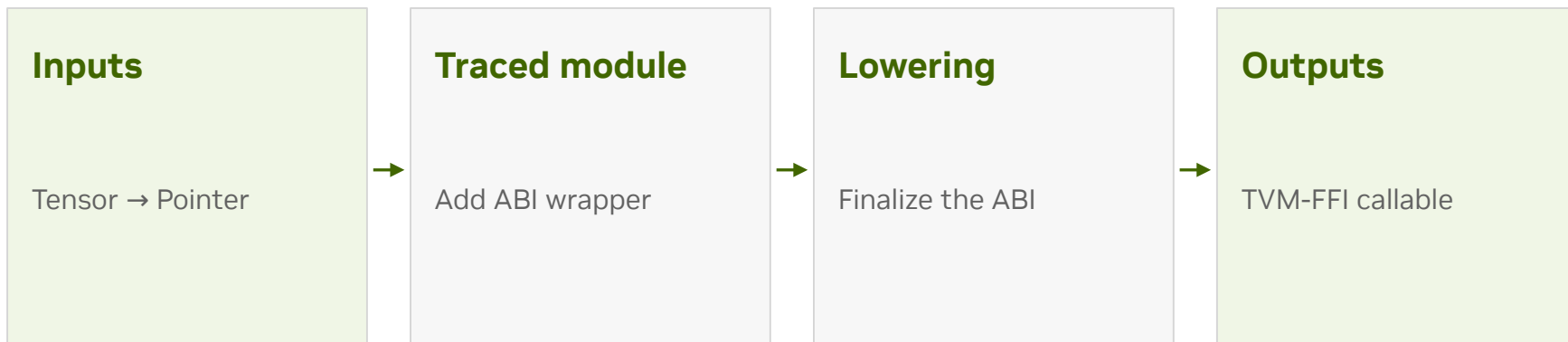
for_ · if_ · while_ · yield_

7 · Bring host values to the DSL



Pointer ABI: contiguous buffers · dimensions passed separately

Adapters work in both directions



AdapterPlugin · register / attach / after_lowering / wrap

8 · @kernel comes from a decorator plugin

GPU plugin + kernel · excerpt

```
plugins = Plugins(  
    # ... selected roles ...  
    decorators=[gpu.Kernels(  
        chip_option="cubin-chip")],  
)  
  
@m.kernel  
def add_one(p: m.Pointer[m.Float32]):  
    i = m.thread_idx()[0]  
    p[i] = p[i] + 1.0
```

Core decorator

@jit

Decorator family

Adds @kernel + launcher

Launch is an explicit operation

Python · implementation excerpt

```
@m.jit
def run(p: m.Pointer[m.Float32]):
    add_one(p).launch(
        m.LaunchConfig(
            grid=[1], block=[32]))

# Caller owns a 32-element CUDA buffer.
run(device_tensor)
```

Prepare → launch

gpu.func → gpu.launch_func

Guard + validate

Grid size · buffer kind

9 · The DSL owns the pass pipeline

Owner	Responsibility
Sub-DSL	pipeline() · register_dialects(ctx)
CompilerPlugin	compile · jit · load
FuncEntryPlugin	Entry op · results ABI
ExecutionEngine compiler	LLVM lowering + packed ctypes call

Compile now. Call later.

Python · implementation excerpt

```
@m.jit
def affine(x: m.Int32, scale, bias):
    return x * scale + bias

compiled = m.compile(affine, 7, 3, 4)
result = compiled(7, 3, 4)
int(result) # 25
```

m.compile

Specialize + compile

Compiled callable

Invoke later

10 · Cache compilation work

Cache	What is reused
In memory	Traced-module key → compiled callable
On disk	Lowered bytecode + CRC32
Configuration	CACHE_DIR · DISABLE_FILE_CACHING
Observation	cache_hits · file_cache_hits

11 · Generate IR and stop

IR-only DSL · setup excerpt

```
class RecipeDSL(BaseDSL):  
    plugins = Plugins(  
        type_ops=TypeOps(scalars=arith),  
        func_entry=func.Entry(),  
        compiler=None,  
    )  
  
# Or use an existing DSL:  
# MLIR_DSL_DRYRUN=1 MLIR_DSL_PRINT_IR=1
```

Compiler is optional

Trace result only

Useful for

Tests · recipe consumers

12 · Errors should point to Python

Python · implementation excerpt

```
@m.jit
def bad(x: m.Int32):
    return int(x)

# PHASE_REQUIRES_CONSTANT
# Runtime value used as Python value
```

Stable codes

Catalogue + suggestions

Useful context

Source frame + DSL message

13 · Locations and remarks

Mechanism	What you see
DEBUGINFO	Python traceback → MLIR locations
REMARKS	Structured trace + compiler feedback
Source frames	Diagnostics point back to the function
Compiler role	Owns the remark session

14 · See what the compiler does

<PREFIX>_*	Purpose
PRINT_IR · DRYRUN	Inspect the traced program
PROFILE_COMPILER	Time compilation phases
JIT_TIME_PROFILING	Measure JIT work
LOG_LEVEL	Control logging

Four roles. Two families.

Eight shipped plugins. Op modules are not plugins.

At most one per role

Type operations

Types + operators

TypeOps

Function entry

Entry + results

func.Entry

AST preprocessor

Syntax capture

scf.ASTPreprocessor

Compiler

Lower + invoke

ExecutionEngine compiler

Any number

Decorators

Add @kernel

gpu.Kernels

Adapters

Inputs + exports

PyTorch · DLPack · TVM-FFI

Shared core

Tracing

Types

Diagnostics

Cache

15 · One subclass. One Plugins record.

Downstream DSL · implementation excerpt

```
class MyDSL(BaseDSL):  
    plugins = Plugins(  
        type_ops=TypeOps(  
            scalars=arith, vectors=vector, memory=llvm),  
        func_entry=func.Entry(),  
        ast_preprocessor=scf.ASTPreprocessor(),  
        compiler=execution_engine.Compiler(),  
        decorators=[gpu.Kernels()],  
        adapters=[pytorch.PyTorchPlugin()],  
    )
```

Select roles

One implementation per role

Extend families

Any number of plugins

Own the namespace and lowering

Downstream DSL · methods

```
def __init__(self):
    super().__init__(
        name="MY_DSL",
        dsl_package_name=["mydsl"],
        preprocess=True)

def register_dialects(self, ctx):
    mydialect.register_dialect(ctx)
```

Downstream DSL · pipeline excerpt

```
def pipeline(self):
    return [
        "convert-mydialect-to-llvm",
        "convert-scf-to-cf",
        "convert-cf-to-llvm",
        "convert-arith-to-llvm",
        "convert-func-to-llvm",
        "reconcile-unrealized-casts",
    ]
```

Reuse a composition; change one field

Python · implementation excerpt

```
from dataclasses import replace

class CpuOnly(m.MlirTestDSL):
    plugins = replace(
        m.MlirTestDSL.plugins,
        decorators=(),
        adapters=(),
    )
```

Keep the roles

Types · entry · AST · compiler

Change the families

Drop GPU + optional adapters

Every plugin has a small shared lifecycle

Shared hook	Meaning
available()	Probe optional support
install(dsl)	Bind this instance
register_dialects(ctx)	Register owned dialects
shared_libs()	Provide runtime libraries

One program: Python recipe + staged loop

Python · executed

```
@m.jit
def fused(arr: m.Pointer[m.Float32],
         ops):
    for i in m.range(4):
        x = arr[i]
        for op in ops.values():
            x = op.apply(x)
        arr[i] = x

# [0, 1, 2, 3] → [6, 9, 12, 15]
```

MLIR · loop body

```
%p = llvm.getelementptr %arr[%i]
      : (!llvm.ptr, i32) → !llvm.ptr, f32
%x = llvm.load %p : !llvm.ptr → f32
%a = arith.addf %x, %two : f32
%r = arith.mulf %a, %three : f32
llvm.store %r, %p : f32, !llvm.ptr
scf.yield
```

Staged array loop · Meta recipe loop

From loop carries to richer mutation

Aliasing · mutable state · PyIR

Amir's section

Advanced mutation tracking

Amir supplies this section.

Return to Guray for a target example.

Tile IR: choose target implementations

Downstream port sketch · proposed API

```
class TileDSL(BaseDSL):  
    plugins = Plugins(  
        type_ops=tile.TypeOps(),  
        func_entry=tile.Entry(),  
        ast_preprocessor=tile.ASTPreprocessor(),  
        compiler=tile.Compiler(),  
        decorators=[tile.Kernels()],  
        adapters=[dlpack.DlpackPlugin()],  
    )
```

Replace target roles

Types · entry · capture · compiler

Reuse infrastructure

Cache · diagnostics · adapters

Tile IR: one type-ops role

Python value	Proposed Tile representation
Int32 / Float32	Rank-0 tile
Vector	Rank-1 tile
Pointer[T]	Tile of pointers
+ · load · store	Target arithmetic + memory ops

Tile IR: choose its control-flow model

Downstream port sketch · proposed API

```
class TileAST(ASTPreprocessorPlugin):  
    preprocessor_class = TilePreprocessor  
  
    def executors(self, dsl):  
        return tile_region_builders(dsl)  
  
# In the Plugins record:  
ast_preprocessor=TileAST()
```

Reuse capture where it fits

Python syntax → target regions

Own target semantics

Early exits · carried values

Tile IR: keep a small user surface

Downstream port sketch · user API

```
@tile.kernel
def add(a: tile.Tensor,
        b: tile.Tensor,
        out: tile.Tensor):
    x = tile.load(a)
    y = tile.load(b)
    tile.store(out, x + y)
```

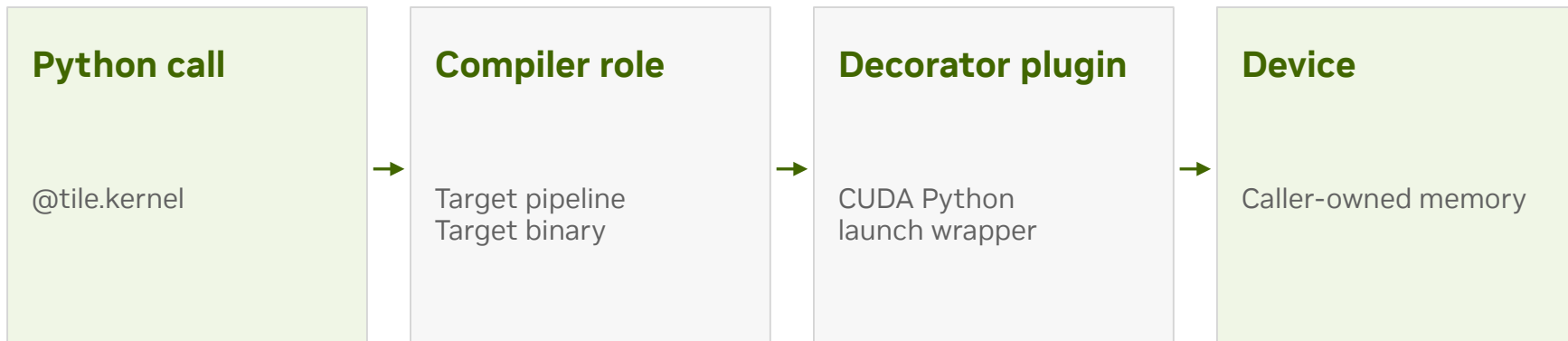
Proposed plugin ownership

```
tile.TypeOps
    types + operations

tile.Kernels
    entry + launch

tile.Compiler
    binary + runtime
```

Tile IR: the target owns launch



Port sketch · reusable core, target-owned execution